

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<think>

- The anatomical snuffbox is a specific area in the anatomical region of the hand .

- The radial artery runs through the anatomical snuffbox .

- A stab wound in this region is most likely to damage the radial artery .

- The ulnar artery is not in this region , and the princeps pollicis artery is a branch of the radial artery .

- The brachial artery is in the upper arm , not the hand .

- Therefore , the most likely damaged artery is the radial artery .

- The question is asking for the artery most likely responsible for excessive bleeding in the given scenario .

- The scenario describes a wound in the anatomical snuffbox , which is innervated by the radial artery .

- Hence , the radial artery is the most likely damaged artery .

- The Princeps pollicis artery is a branch of the radial artery , not a separate artery .

- The ulnar artery and brachial artery are not in the same anatomical region .

- Therefore , the radial artery is the correct answer .

- The question is asking for the artery most likely responsible for excessive bleeding .

- The radial artery is the most likely artery to be damaged in this scenario .

- The radial artery is the correct answer .

- The radial artery is the correct answer .

</think>

<answer>A</answer>